
IPv6

Conventions for Representing IPv6 Addresses

- 128-bit
- 32 hexadecimal numbers, organized into 8 quartets of 4 hex digits separated by a colon

Hex	Binary	Hex	Binary
0	0000	8	1000
1	0001	9	1001
2	0010	10	1010
3	0011	11	1011
4	0100	12	1100
5	0101	13	1101
6	0110	14	1110
7	0111	15	1111

Example

2340:1111:AAAA:0001:1234:5678:9ABC:0001

Conventions for Representing IPv6 Addresses

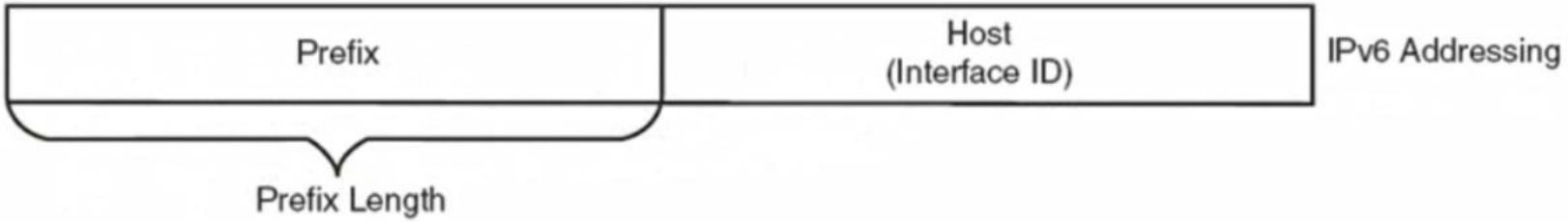
To make things a little easier:

- Omit the leading 0s in any given quartet.
- Represent one or more consecutive quartets of all hex 0s with "::" but only for one such occurrence in a given address.

Example

FE00:0000:0000:0001:0000:0000:0000:0056

Conventions for Representing IPv6 Addresses



Example

2000:1234:5678:9ABC:1234:5678:9ABC:1111/64

IPv6 Assignment

Term	Assignment	Example from Chapter 16
Registry prefix	By IANA to an RIR	2340::/12
ISP prefix	By an RIR to an ISP[1]	2340:1111/32
Site prefix or global routing prefix	By an ISP or registry to a customer (site)	2340:1111:AAAA/48
Subnet prefix	By an Enterprise engineer for each individual link	2340:1111:AAAA:0001/64

TCP/IP Configuration Assignment Overview

Method	Dynamic or static	Prefix and length learned from...	Host learned from...	Default router learned from...	DNS addresses learned from...
Stateful DHCP	Dynamic	DHCP Server	DHCP Server	Router, using NDP	(Stateful) DHCP Server
Stateless autoconfig	Dynamic	Router, using NDP	Derived from MAC	Router, using NDP	Stateless DHCP
Static configuration	Static	Local config	Local config	Router, using NDP	Stateless DHCP
Static config with EUI-64	Static	Local config	Derived from MAC	Router, using NDP	Stateless DHCP

State Full DHCP

Method	Dynamic or static	Prefix and length learned from...	Host learned from...	Default router learned from...	DNS addresses learned from...
Stateful DHCP	Dynamic	DHCP Server	DHCP Server	Router, using NDP	(Stateful) DHCP Server

- The names and formats of the actual DHCP messages have changed quite a bit from IPv4 to IPv6, so DHCPv4 and DHCPv6 actually differ in detail, but the basic process remains the same.

- IPv4 hosts send IP broadcasts to find DHCP servers, whereas IPv6 hosts send IPv6 multicasts.

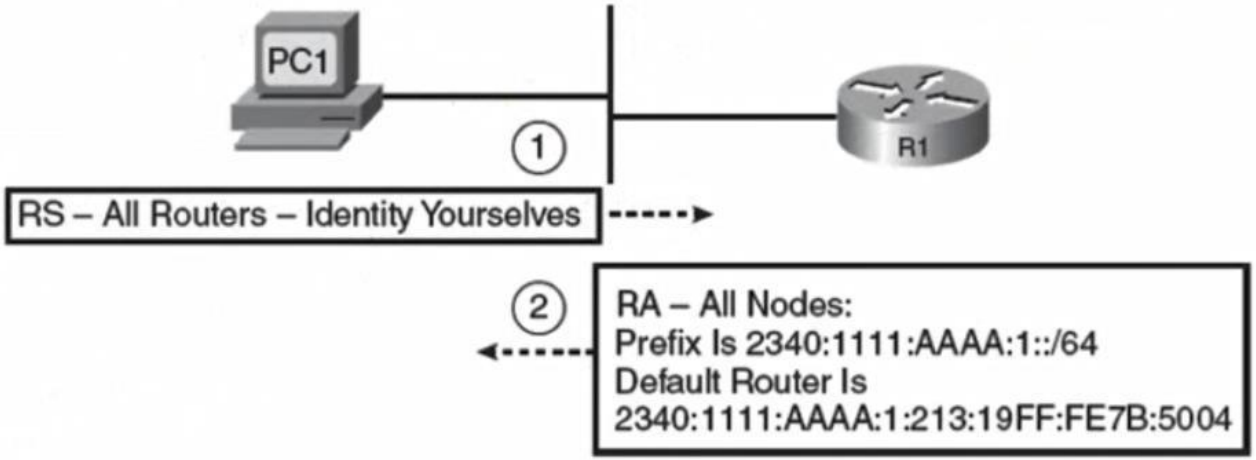
State Less Auto Configuration

Method	Dynamic or static	Prefix and length learned from...	Host learned from...	Default router learned from...	DNS addresses learned from...
Stateless autoconfig	Dynamic	Router, using NDP	Derived from MAC	Router, using NDP	Stateless DHCP

- **IPv6 Neighbor Discovery Protocol (NDP)**
  - Prefix
  - Prefix length
  - Default router
- **EUI-64**
  - Interface ID
- **Stateless DHCP**
  - DNS IPv6 addresses

State Less Auto Configuration

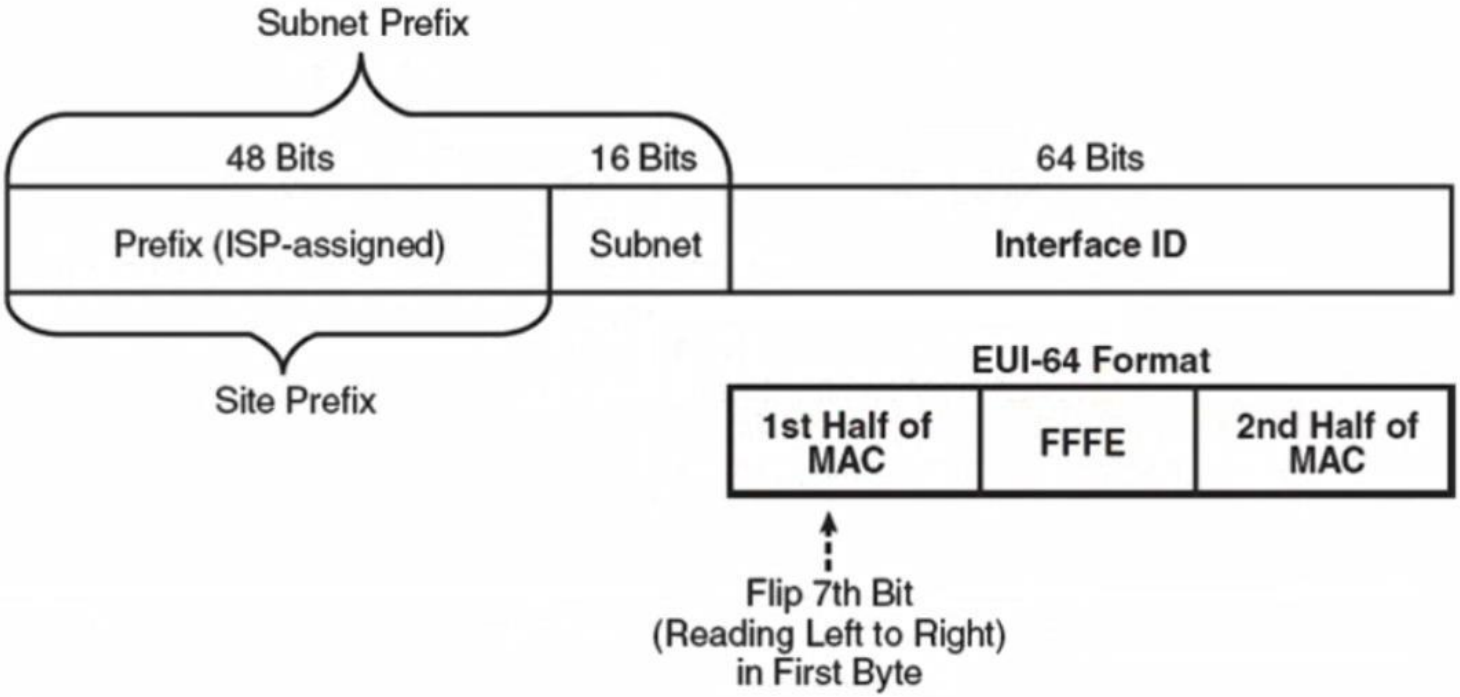
IPv6 Neighbor Discovery Protocol (NDP)



Message	RS	RA
Multicast destination	FF02::2	FF02::1
Meaning of Multicast address	All routers on this link	All IPv6 nodes on this link

State Less Auto Configuration

EUI-64



Static IPv6 Address Configuration

Method	Dynamic or static	Prefix and length learned from...	Host learned from...	Default router learned from...	DNS addresses learned from...
Static configuration	Static	Local config	Local config	Router, using NDP	Stateless DHCP
Static config with EUI-64	Static	Local config	Derived from MAC	Router, using NDP	Stateless DHCP

IPv6 Unicast Address Summary

Type of Address	Purpose	Prefix	Easily Seen Hex Prefix(es)
Global unicast	Unicast packets sent through the public Internet	2000::/3	2 or 3
Unique local	Unicast packets inside one organization	FD00::/8	FD
Link local	Packets sent in the local subnet	FE80::/10	FE8
Site local	Deprecated; originally meant to be used like private IPv4 addresses	FECO::/10	FEC, FED, FEE, FEF

IPv6 Unicast Address Summary

Unspecified	An address used when a host has no usable IPv6 address	::/128	N/A
Loopback	Used for software testing, like IPv4's 127.0.0.1	::1/128	N/A

IPv6 RFCs define the FE80::/10 prefix, which technically means that the first three hex digits could be FE8, FE9, FEA, or FEB. However, bit positions 11-64 of link local addresses should be 0, so in practice, link local addresses should always begin with FE80.

IPv6 Multicast Address Summary

Purpose	IPv6 Address	IPv4 Equivalent
All IPv6 nodes on the link	FF02::1	subnet broadcast address
All IPv6 routers on the link	FF02::2	N/A
OSPF messages	FF02::5, FF02::6	224.0.0.5, 224.0.0.6
RIP-2 messages	FF02::9	224.0.0.9
EIGRP messages	FF02::A	224.0.0.10
DHCP relay agents (routers that forward to the DHCP server)	FF02:1:2	N/A
DHCP servers (site scope)	FF05::1:3	N/A
All NTP servers (site scope)	FF05::101	N/A

IPv6 Configuration Summary

Command	Description
ipv6 address *address/length*	Static configuration of the entire IPv6 unicast address.
ipv6 address *prefix/length* eui-64	Static configuration of the first 64 address bits; the router derives the last 64 bits with EUI-64.
ipv6 address autoconfig	Router uses stateless autoconfig to find address.
ipv6 address dhcp	Router uses stateful DHCP to find address.
ipv6 unnumbered *interface-type number*	Uses the same IPv6 unicast address as the referenced interface.
ipv6 enable	Enables IPv6 on the interface, but results in only a link local address.
ipv6 address *address* link-local	Overrides the automatically created link local address. The configured value must conform to the FE80::/10 prefix.
ipv6 address *address/length* anycast	Designates that the unicast address is an anycast.

IPv6 Configuration

Note

IPv4 and IPv6 Dual Stack

Configuring RIPng

Router(config)# ipv6 unicast-routing
Router(config)# ipv6 router rip <*name*>

Router(config)# interface type *mod/num*
Router(config-if)# ipv6 rip <*name*> enable

Configuring EIGRP for IPv6

Router(config)# ipv6 unicast-routing
Router(config)# ipv6 router eigrp <*as#*>
Router(config-router)# no shutdown
Router(config-router)# eigrp router-id <*RID*>

Router(config)# interface type *mod/num*
Router(config-if)# ipv6 eigrp <*as#*>

OSPF Version 3 Configuration

Router(config)# ipv6 unicast-routing
Router(config)# ipv6 router ospf <*process-ID*>
Router(config-router)# router-id <*RID*>

Router(config)# interface type *mod/num*
Router(config-if)# ipv6 ospf <*process-ID*> **area** <*area#*>